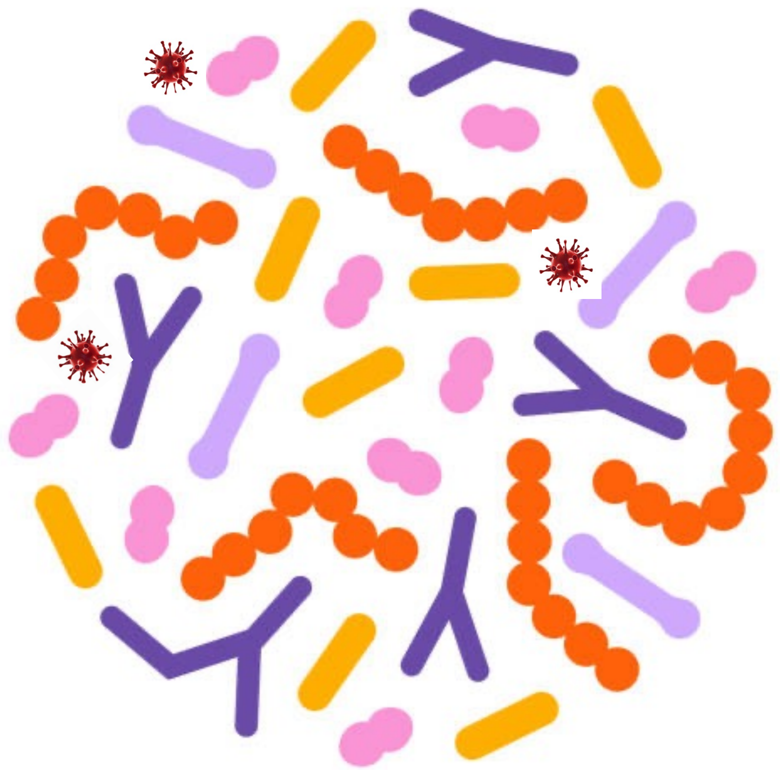

Processing of shotgun metagenomics samples

- nf-core/taxprofiler together with Metaphlan v4 database
- Genome-level marker gene based taxonomic classifier

- Data Science platform
- BRIGHT
- Spring Data-type Courses
- 6th May 2026

Metagenomic insights into the prokaryotic communities of heavy metal-contaminated hypersaline soils

Cristina Galisteo [a], Fernando Puente-Sánchez [b], Rafael R. de la Haba [a], Stefan Bertilsson [b], Cristina Sánchez-Porro [a], Antonio Ventosa [a]

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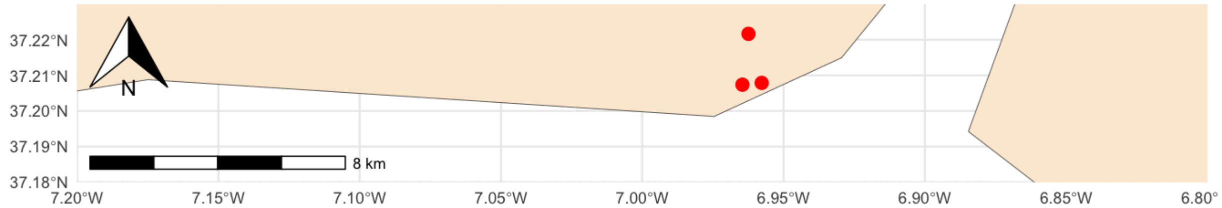
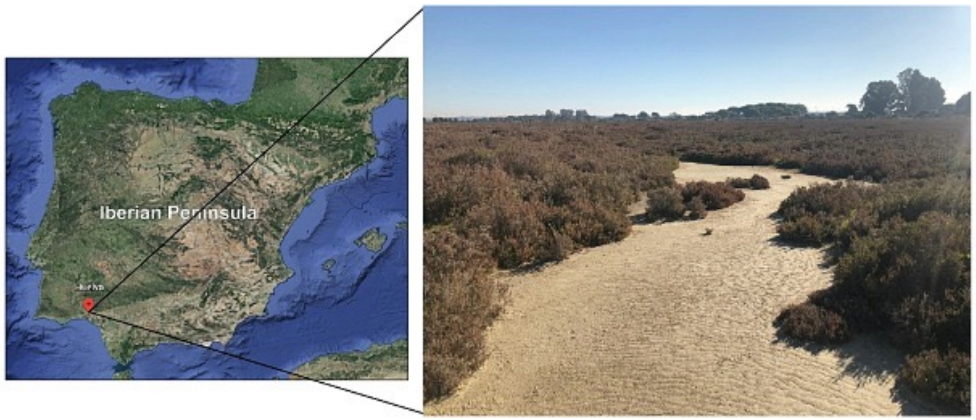
 Open access

- Saline soils and their microbial communities have recently been studied in response to ongoing desertification of agricultural soils caused by anthropogenic impacts and climate change.
- In this study, they describe the prokaryotic microbiota of hypersaline soils in the Odiel Saltmarshes Natural Area of Southwest Spain.
- This region has been strongly affected by mining and industrial activity and feature high levels of certain heavy metals.
- **18** shotgun metagenomes sequenced with Illumina NovaSeq from three different areas in 2020 and 2021.

Sampling sites



- Three sampling sites (1, 2 and 3) were selected in order to obtain a representative subset of the prokaryotic community of the soils from the Odiel Saltmarshes Natural Area



Sample table with some metadata:

- **Sampling dates (2)**
- **Areas (3)**
- Coordinates
- Sample names (n = 18)

Sampling	Date	Area	Samples	Coordinates
M2	July 2020	1	M2_1A	37°12′26.6″N 6°57′52.5″W
			M2_1B	
			M2_1C	
		2	M2_2A	37°12′28.4″N 6°57′27.9″W
			M2_2B	
			M2_2C	
		3	M2_3A	37°13′18.0″N 6°57′44.8″W
			M2_3B	
			M2_3C	
M3	June 2021	1	M3_1A	37°12′26.6″N 6°57′52.5″W
			M3_1B	
			M3_1C	
		2	M3_2A	37°12′28.4″N 6°57′27.9″W
			M3_2B	
			M3_2C	
		3	M3_3A	37°13′18.0″N 6°57′44.8″W
			M3_3B	
			M3_3C	

Processing metagenomics data – a fastq file

A00180 → instrument name
10 → run id
H7NK5DMXX → flowcell id
1 → flowcell lane
1101 → tile number within the flow cell line
16821 → x-coordinate of the cluster within the tile
1000 → y-coordinate of the cluster within the tile
SPACE
1 → member of the pair (if paired-end)
N → N if read is not filtered, Y if it is filtered
0 → when none of the control bit are on
GGACTT, GTCGTTCG → barcodes are used for demultiplexing and assign reads to the correct sample

Four lines for each read:

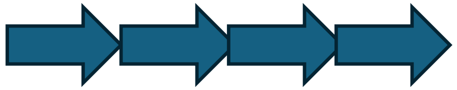
1. identifier (starting with @)
2. Read sequence
3. a line starting with + (sometimes id gets repeated)
4. Phred quality scores representing the base-caller's confidence – equal in length to the read sequence

```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTCG
TGCAGCAGCTAATGAGGAACCACTTCCTCCCTCCAGCCGCTCTAAATACCTCAGAACAATAGGATCATCATAATAATCCCCTAGTCTGAACTG
+
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAAGAAGAAAGCACAAGTATTTACGCCTATCCTTCATATTTTCCGCAAGGTAACTATCTCGGTTTCATATCGAGATTTATATAGAATCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-FFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGGAAGTTATGCAGGGGCATCCTGTATTATTAAATAGAGCACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG
+
FF-88F8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-F-FFFFFFF
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAGTACATCAACAAAAGAAACCTAACTGCCCTACCGGCAAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCCAGTCAATATAAGG
+
8F888FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-FFFFFFFFFFFFFFF-FFFFFFFFFFFFFFF-FFFFFFFFFF
```

Repeating steps too many times? – find yourself a pipeline!

```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTCG
TGCAGCAGCTAATGAGGAACCACTTCCTCCCTCCAGCCGCTCTAAATACCTCAGAACAATAGGATCATCATAATAATCCCCTAGTCTGAACTG
+
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAAGAAGAAAGCACAAGTATTTACGCCTATCCTTCATATTTTCCGCAAGGTAACTATCTCGGTTTCATATCGAGATTTATATAGAATCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-FFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGGAAGTTATGCAGGGGCATCCTGTATTATTAAATAGAGCACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG
+
FF-88F8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-F-FFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAGTACATCAACAAAAGAAACCTAACTGCCCTACCGGCAAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCCAGTCAATATAAGG
+
8F888FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-FFFFFFFFFFFFFFF-FFFFFFFFFFFFFFF-FFFFFFFFFF
```

Series of steps in an specific order



Abundance table for downstream analysis

OTU	Sample 1	Sample 2	Sample 3
A	0.60	0	0.70
B	0.24	0.20	0.10
C	0.10	0	0
D	0.05	0	0
E	0.01	0	0
F	0	0.80	0.20
Total	1.0	1.0	1.0

- Different steps to run manually in an order (QC, host removal, mapping to a database, etc...)
- Workflow that contains all the processes needed and have them chained

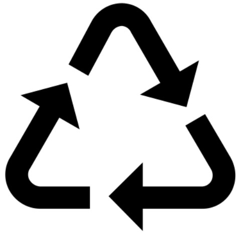
nextflow

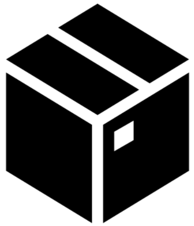
Nextflow is a language to build workflows and a software to orchestrate workflows execution.

- Strong community behind with various bioinformaticians joining forces to build each of the pipelines
- Support for other technologies

Nextflow’s core features

- **Reproducible** between runs
  - integration with code management tools
  - all packages downloaded, organized in containers, and control over computing environment
- **Portable** between systems
  - you can write the code in your laptop and can run everywhere (HPC, cloud)
  - works with most of computing environments
- **Scalable**
  - it can run for 10 samples on your laptop or thousands in the cloud
- **Integration** of existing tools, systems, and industry standards

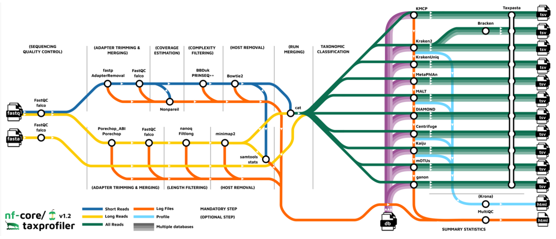


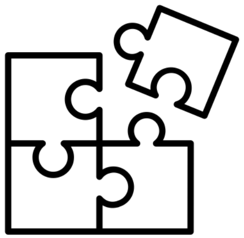


nf-core

A global community effort to collect a curated set of open-source analysis pipelines built using Nextflow.
https://nf-co.re/

https://www.nature.com/articles/s41587-020-0439-x





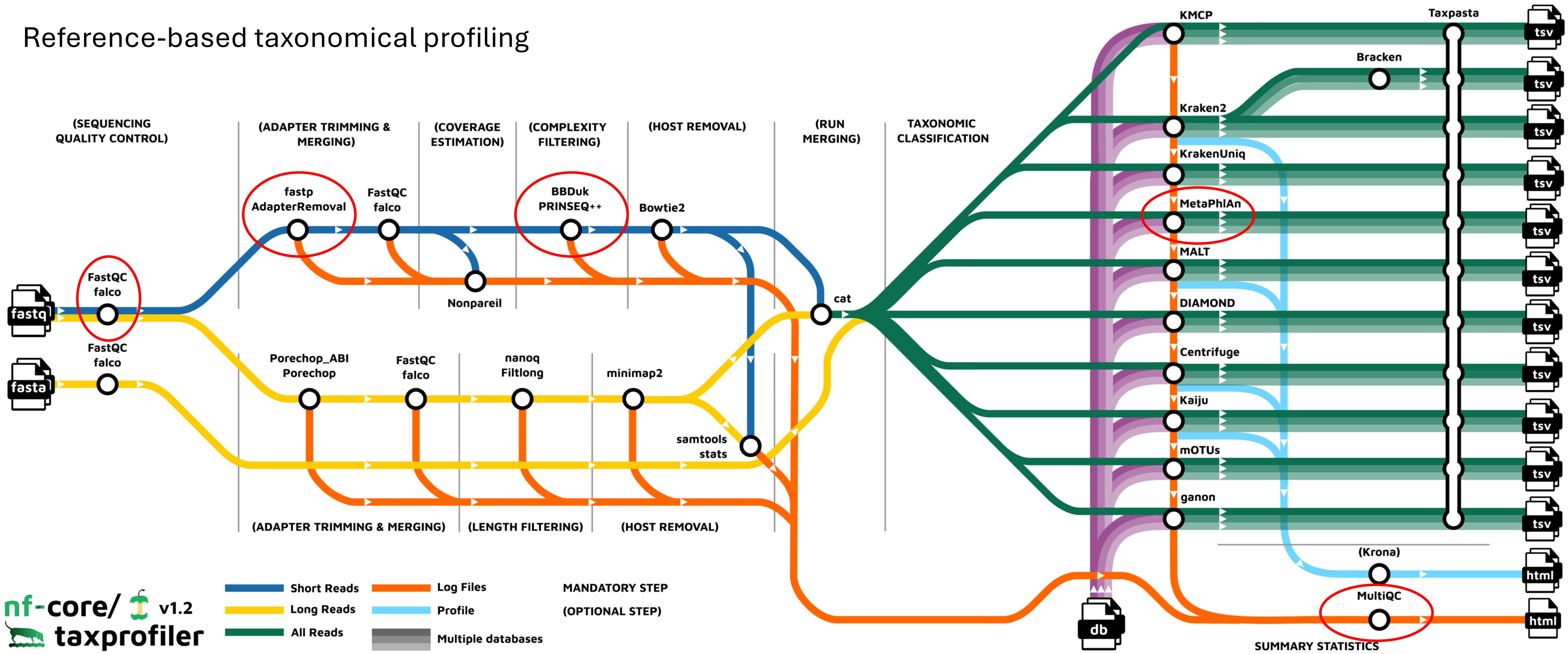
149 **pipelines** available (2nd May 2026) for 149 different bioinformatic analysis

Modules: 1,811 reusable components that can be integrated into pipelines

Subworkflows: 117 pre-assembled combinations of modules aimed at streamlining commonly used workflows

nf-core/taxprofiler

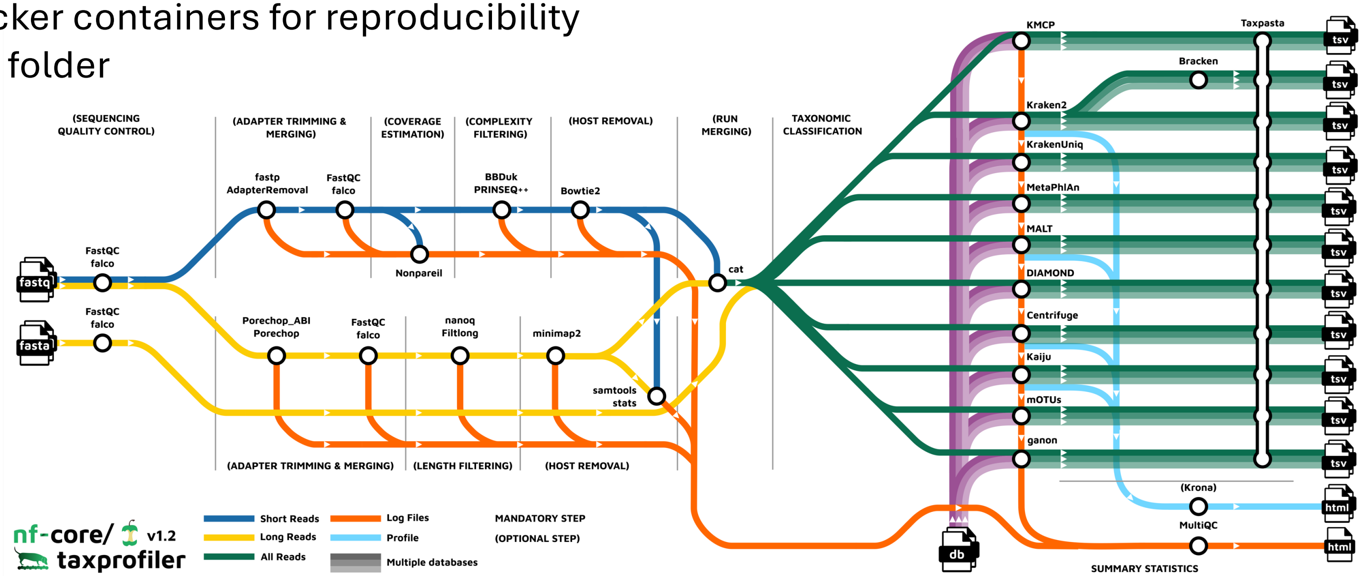
Reference-based taxonomical profiling



Command line to run the pipeline

➢ nextflow run nf-core/taxprofiler -profile test,docker --outdir processed_data

nf-core/taxprofiler → nf-core pipeline downloaded from here: https://github.com/nf-core/taxprofiler/tree/2.0.0
“test” → running for demonstration few reduced samples, each pipeline has a test dataset (in a github repo, so we do not have to specify the input files this time)
“docker” → using Docker containers for reproducibility
--outdir → our results folder



Wanna know more about Nextflow: https://biosustain.github.io/dsp_nextflow_training/